Neural Networks Training

Introduction

Slides are partially adapted from notes by William Kaiser, Vincent Cheung, and Kevin Cannons

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Recall: Why use neural networks?

Ability to learn

- Can figure out how to perform their function on their own
- Determine their function based only upon sample inputs

Ability to generalize

- Produce reasonable outputs for inputs it has not been taught how to deal with

How do neural networks learn?

- In other words, how to obtain the weights (and biases) to approximate the function of a system?
- **Training** is the act of presenting the network with some sample data and modifying the weights (and biases) to better approximate the desired function.

Supervised Training

- Supplies the neural network with inputs and the desired outputs
- Response of the network to the inputs is measured.
- The weights (and biases) are modified to reduce the difference between the actual and desired outputs.
- The process of providing the network with an input (or a batch of inputs) and updating the network's weights (and biases) is one iteration.
- Typically, many iterations are required to train the neural network.

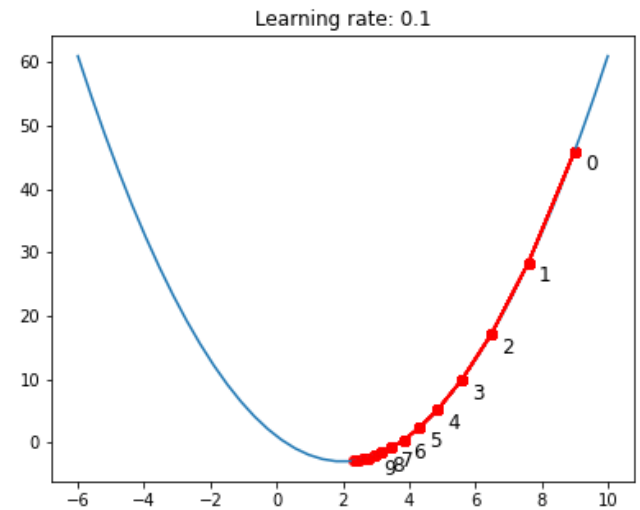
Gradient Descent

- **An iterative optimization algorithm used to find a local minimum/maximum of a given function.**
- **Commonly used in machine learning and deep learning to minimize a cost/loss function.**
- **It iteratively calculates the next point using gradient at the current position, then scales it (by a learning rate) and subtracts obtained value from the current position to make a step.**
  - Subtract when to minimize a function, add when to maximize a function.

Gradient Descent (cont'd)

Algorithm:

- $x_{n+1} = x_n - \alpha f'(x_n)$, α is the learning rate



Example:

- $f(x) = x^2 - 4x + 1$
- $f'(x) = 2x - 4$
- Assume α = 0.1, starting point is $x_0 = 9$
  - $x_1 = x_0 - \alpha f'(x_0) = 9 - 0.1 * (2 * 9 - 4) = 7.6$
  - $x_2 = x_1 - \alpha f'(x_1) = 7.6 - 0.1 * (2 * 7.6 - 4) = 6.48$
  - $x_3 = x_2 - \alpha f'(x_2) = 6.48 - 0.1 * (2 * 6.48 - 4) = 5.584$
  - …
  - Eventually converge to minimum at x = 2

Question

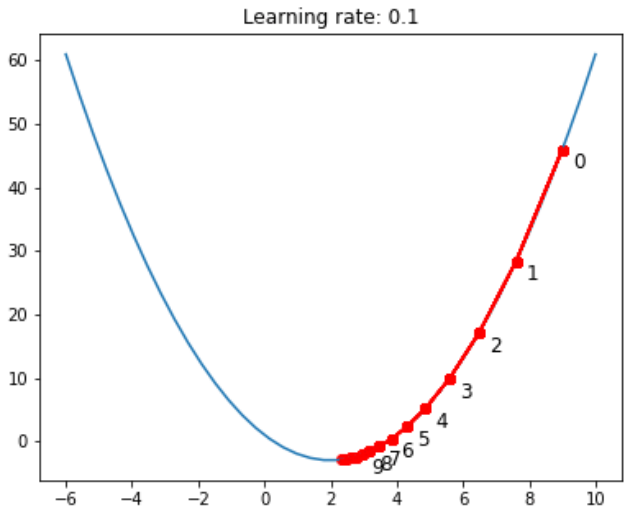
Continue from the previous example, what is x_4?

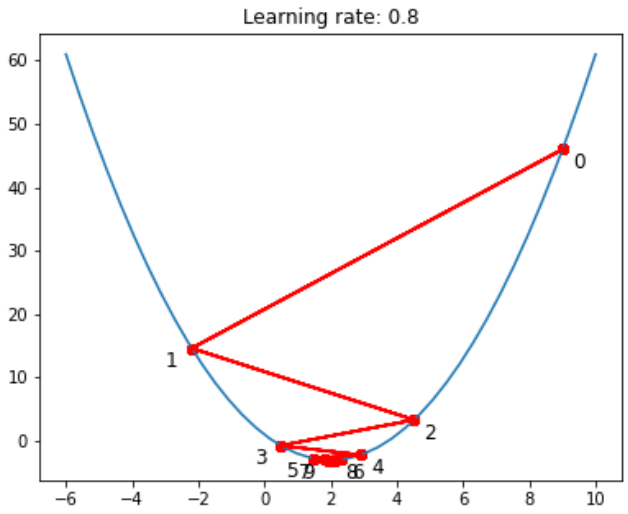
- $f(x) = x^2 - 4x + 1$
- $f'(x) = 2x - 4$
- Assume α = 0.1, starting point is $x_0 = 9$
  - $x_1 = x_0 - \alpha f'(x_0) = 9 - 0.1 * (2 * 9 - 4) = 7.6$
  - $x_2 = x_1 - \alpha f'(x_1) = 7.6 - 0.1 * (2 * 7.6 - 4) = 6.48$
  - $x_3 = x_2 - \alpha f'(x_2) = 6.48 - 0.1 * (2 * 6.48 - 4) = 5.584$
  - $x_4 = ?$

Gradient Descent (cont'd)

The learning rate is important

- Too small: converge very slowly
- Too large: overshoot, may not converge





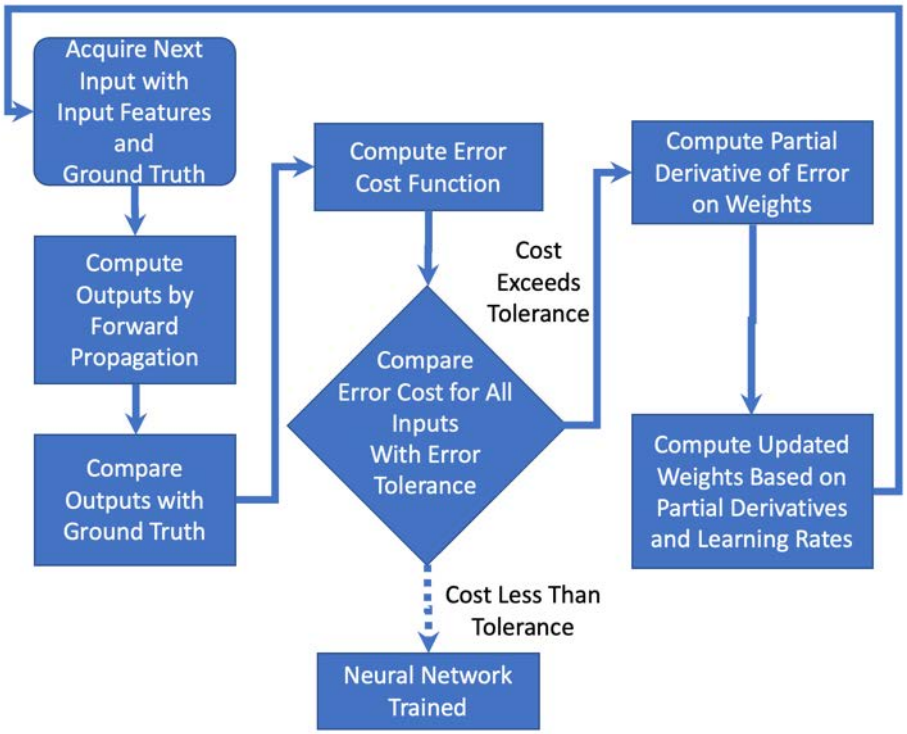
How do MLP neural networks learn?

The most common method of obtaining the many weights (and biases) in the network is backpropagation.

The basic backpropagation algorithm is based on minimizing the error of the network using the derivatives of the error function.

- Gradient descent algorithm
  - $w_{new} = w_{old} - \alpha\, \partial E/\partial w_{old}$, α is the learning rate, w is weight, E is error.

Gradient Descent Training Algorithm



Training Set vs. Test Set

The available dataset including network inputs and outputs (ground truth) is usually split into two sets:

- Training set: used to train the neural network model.
- Test set: used to get an unbiased estimate of the final model performance.

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